Pediatric Nephrology (2026) 41:45–59
https://doi.org/10.1007/s00467-025-06771-9

EDUCATIONAL REVIEW

Impact of maternal health on neonatal and long-term kidney outcomes

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Received: 7 November 2024 / Revised: 3 March 2025 / Accepted: 24 March 2025 / Published online: 3 June 2025
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Abstract

The increasing prevalence of non-communicable diseases like chronic kidney disease signals the need for a deeper understanding of the impact of the intra-uterine milieu on developmental programming and its impact on health outcomes through the lifespan of an individual. Maternal health in the pre-gestational and gestational phases, including nutrition, exposure to drugs, environmental toxins, infectious and non-infectious diseases, and socio-economic conditions influence the overall development of the fetus as well as the fetal kidney. The small, vulnerable newborn, born from an adverse developmental environment, is predisposed to low nephron number and is at risk for acute kidney injury in the neonatal period. Developmental programming has far-reaching consequences, including a higher risk for cardio-kidney-metabolic diseases, including hypertension and chronic kidney disease, and pregnancy complications, which perpetuates an intergenerational cycle of non-communicable disease risk. This risk can be mitigated by optimizing the care of individuals in the reproductive age group, identifying high-risk pregnancies early, and providing optimal treatment and monitoring. Care of the small vulnerable neonate includes prevention of acute kidney injury and life-long surveillance and modulation of cardio-kidney-metabolic risk. The review focuses on highlighting the influence of maternal health in the pre-gestational and gestational phases on kidney health from the neonatal period to adulthood.

Keywords Preeclampsia · Nephron number · Small vulnerable newborn · Chronic kidney disease · Neonatal acute kidney injury · Low birth weight · Preterm birth

Introduction

In 2020 it was estimated that 13.4 million children were born preterm (before the 37th week of gestation) worldwide, while an even higher number, 21.9 million children, were born small for gestational age (SGA, birth weight below 10th centile and/or low birth weight (LBW), < 2500 g) [1]. The term Small Vulnerable Newborns (SVN), totaling 35.3 million infants in 2020, reflects a conceptual framework that brings together preterm births, SGA, and LBW. These infants are collectively identified as vulnerable because they experience adversities during the intrauterine period, which lead to stressed growth processes for the fetus and contribute to a high risk of neonatal death and longer-term risk

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of non-communicable diseases (NCDs), including kidney disease [2, 3].

Over the last few decades, NCDs have replaced communicable diseases as the leading causes of premature mortality across the world [4]. The global prevalence and mortality from chronic kidney disease (CKD), one of the important NCDs, is projected to continue to increase in the next few decades [5]. The intrauterine milieu, determined by the mother's health and her own birth circumstances, nutrition, exposures to drugs, infections, and toxins, her socio-economic situation, and access to antenatal care has a great impact on the overall development of the fetus, and in particular on the fetal kidney [6]. This process has been termed developmental programming. CKD and its most common etiological precursors, diabetes and hypertension, have links to developmental programming in fetal life [6]. The associations between developmental programming, as reflected by prematurity, LBW and growth restriction at birth, and several NCDs in later life i.e. obesity, glucose intolerance, diabetes, hypertension, cardiovascular disease, preeclampsia and CKD, are now well recognized [6–12].

In this review, we explore the influence of maternal factors in the pre-gestational and gestational periods on the development of the fetal kidney and its impact on kidney health through the lifespan of the offspring.

Maternal health

Maternal health and well-being, as well as access to antenatal care (ANC), are closely interconnected with neonatal outcomes and the life-long health of the offspring [13, 14]. In a recent study from China, SGA births were associated with lower maternal educational level, lower age, higher parity, fewer prenatal visits, and pre-existing maternal diseases [15]. Here, we review maternal factors associated with SVNs and thereby with kidney development and risk of later-life kidney disease in the offspring, which may be present pre-gestation and during gestation. We follow with a discussion on maternal factors which impact neonatal and long- term kidney health.

Pre-gestational risk factors

Maternal age

Both extremes of reproductive age predispose to adverse perinatal outcomes. Pregnancies in individuals under 19 years of age are more prone to preterm labor and LBW [16]. Better education for girls and access to family planning are important to reduce these risks [6], and this is improving globally [17]. Individuals above 35 years of age are more prone to preterm births which are associated with pregnancy-related complications like preeclampsia, and a greater prevalence of chronic disease [18]. Women with pre-existing chronic diseases like CKD are often counseled to delay pregnancy until either their primary disease is under control, or they are established on kidney replacement therapy (preferably kidney transplant). Women with chronic diseases may, therefore, conceive at an older age [19]. In addition, with improved reproductive technologies, rates of conception in people with comorbidities and older women have increased. The risk of delaying pregnancy in such cases must be carefully balanced with the risk of maternal and fetal complications associated with advanced maternal age [20].

Maternal pre-pregnancy weight, nutrition, and pre-pregnancy exposures

Both extremes of maternal weight are associated with neonatal morbidity [21]. Higher maternal BMI increases the risk of macrosomia, premature birth, and neonatal asphyxia [21]. Maternal underweight is associated with LBW and SGA [21]. Deficiencies of maternal folate, vitamin A, and total energy intake impact infant kidney structure and function as measured by kidney volume, proteinuria, and estimated GFR [22, 23]. Maternal smoking or secondhand smoke exposure during pregnancy is associated with lower birthweight and adverse pregnancy outcomes [24]. Maternal smoking also increases the risk of childhood obesity and metabolic syndrome, which may compound the risk of CKD when nephron numbers are lower [25]. Optimization of maternal nutrition and exposures pre-pregnancy are important to support fetal kidney development [26].

Maternal health

The presence of maternal comorbidities (hypertension, diabetes, obesity, and cardiovascular diseases) is closely linked with complications during pregnancy, including instrumental delivery and preterm birth [27]. Individuals with pre-existing comorbidities may have significant endothelial dysfunction which may drive placental insufficiency and preeclampsia [28]. As mentioned above, maternal comorbidities may be associated with increased maternal age and use of assisted reproductive techniques. In addition, the frequency of congenital abnormalities of the kidney and urinary tract (CAKUT) are also impacted by maternal health during pregnancy [29, 30]. Women with comorbidities may take medications that require changes pre-conception (see below). Pregnancies in persons with comorbidities should be considered high risk and planned and followed appropriately to optimize both maternal and fetal outcomes.

Pregestational diabetes

The presence of both type 1 and type 2 diabetes and resultant hyperglycemia increases the risk of maternal and fetal complications including preeclampsia, fetal malformations, macrosomia, preterm birth, early pregnancy loss, and perinatal mortality [31]. These risks are directly proportional to the blood glucose levels in the pre-conception period, and tight glycemic control is associated with improved outcomes [32]. Exposure to diabetes in utero is associated with increased risk of albuminuria and CKD in adult offspring, although a co-programmed risk of diabetes may also modulate this [33]. Prospective mothers should be educated about the importance of strict glycemic control in the pre-gestational and gestational periods.

Maternal kidney disease

CKD is seen in about 3% of people of child-bearing age [34]. All stages of CKD (including stage 1) are associated with adverse pregnancy outcomes [35]. The rate of preterm birth in women with CKD is several-fold higher than controls (44% vs. 5%) [35]. The presence of proteinuria > 1 g/day and hypertension are important predictors of poor maternal–fetal outcomes [35].

The rates of conception on peritoneal dialysis are very low and thus a shift to hemodialysis before conception may be considered [36, 37]. Amongst patients on dialysis, reported live birth rates range from 40 to 86%, and 50%—100% of pregnancies result in preterm births [36, 38, 39]. These outcomes have been attributed to a higher incidence of hypertensive disorders of pregnancy and HELLP syndrome [39, 40]. In advanced CKD, the best maternal and fetal outcomes are reported in kidney transplant recipients with live birth rates comparable to the normal population [41, 42]. However, there is a 4 to 6 times higher risk of preeclampsia, preterm birth, and LBW in this population when compared to women with normal kidney function [41, 42]. Pre-conception counselling is therefore required for all women with kidney disease.

Systemic lupus erythematosus

Systemic lupus erythematosus (SLE) affects individuals in the reproductive age group. SLE has profound implications for the mother, with a high risk of preeclampsia, lupus flares, as well as for the neonate in terms of SGA and heart block [43]. Factors associated with a greater risk of preterm birth and preeclampsia in women with SLE include previous lupus nephritis, SLE disease activity, chronic hypertension, and antiphospholipid syndrome [44]. The timing and safety of pregnancy must be very carefully considered in women with SLE.

Maternal infections in pre-gestational period

Infections in the preconception period which remain untreated during pregnancy can have a variety of adverse outcomes on the offspring [45]. Sexually transmitted infections (STIs) may impact fertility, and increase the risk of stillbirths and SVN [46]. Early identification and treatment of infections should be emphasized.

Socio-economic factors and pre-gestational access to healthcare

People with lower socio-economic status (SES) have lower health literacy and poorer access to healthcare than those of higher SES [47, 48]. Mothers with lower SES are more likely to develop gestational diabetes, gestational hypertension, and preeclampsia even after adjusting for ethnicity and baseline health status [48]. These women also experience more delivery-related complications, like premature rupture of membranes and antepartum hemorrhage [49]. All these factors increase the risk of SVN. Importantly, pregnancy is often the first and only time many individuals of low SES seek healthcare. This may lead to chronic conditions remaining undiagnosed and contributing to adverse pregnancy outcomes [50]. The World Health Organization currently recommends at least 4 antenatal visits, which is currently only achieved by around 70% of pregnant women globally [51]. Optimal interventions in antenatal care are discussed further below.

Gestational risk factors

Preeclampsia

Preeclampsia is a syndrome of hypertension in pregnancy, which is characterized by widespread maternal endothelial dysfunction [52]. It has a profound impact on maternal and fetal mortality and morbidity, and accounts for 8−10% of preterm births [53]. Neonates born to pre-eclamptic mothers have increased risks of being SGA (aOR 5.3, CI: 5.1–5.5) and needing resuscitation (aOR 2.6, CI: 2.4–2.7) [54]. These neonates are also at increased risk of a low Apgar score, hypoxic-ischemic encephalopathy, neonatal sepsis, and perinatal death [54]. Long-term outcomes of preeclampsia exposure in the offspring include cardiovascular and metabolic disorders [53]. Also, mothers with preeclampsia are at risk of developing chronic hypertension, cardiovascular and kidney disease, which further compromise the outcomes in subsequent pregnancies [55]. Screening for preeclampsia, appropriate treatment, and early referral for high-risk cases is crucial.

Gestational diabetes

The incidence of gestational diabetes (GDM) is increasing especially in the context of obesity [56]. Uncontrolled, long-standing diabetes with vascular complications affects placental growth resulting in fetal growth restriction [57]. GDM with or without hypertension can be a risk factor for prematurity [58].

GDM also exposes the fetus to short-term complications such as macrosomia, and developmental defects in many organ systems including the genito-urinary system [23, 59]. Increased risks of perinatal asphyxia and renal vein thrombosis are potential mechanisms that increase the risk of neonatal AKI [60].

Offspring exposed to GDM are also at risk of long-term consequences such as obesity, type 2 diabetes mellitus, cardiovascular disease, and CKD [61]. Indeed, population-based data show that exposure to GDM is associated with an odds ratio of 1.54 (95% CI, 1.13 to 2.09) for childhood CKD. In this study, the risk of kidney dysplasia was higher when GDM was also associated with infant LBW, highlighting the co-existence of multiple programming risks in at-risk pregnancies [59]. Early diagnosis, strict glycemic control, and anticipating perinatal complications play a key role in managing mothers with GDM.

Pregnancy-related infections/sepsis

Bacterial vaginosis caused by Group B streptococcus, E. coli, Klebsiella, Enterococcus, and Ureaplasma urealyticum are associated with early preterm births [45]. Maternal Coxsackievirus, Rubella, Human Immunodeficiency Virus (HIV) and Syphilis in the first trimester may lead to congenital anomalies, intrauterine growth restriction (IUGR), and stillbirth [46]. Malaria infection during pregnancy is associated with SVN births and prevention is strongly recommended for pregnant individuals in endemic areas [62].

Urinary tract infections (UTI) are found in up to 8% of pregnancies [63]. Untreated UTI in pregnancy can lead to preterm birth [64]. Long-standing untreated UTI can also predispose to IUGR, preterm labor, and LBW. Ascending infection can frequently lead to pyelonephritis in pregnancy, which may lead to amnionitis and perinatal mortality [65]. Early diagnosis and prompt management of infections during pregnancy can reduce the risk of SVN.

Pregnancy-associated acute kidney injury

In individuals with pregnancy-related AKI, about 35% of infants experience adverse outcomes including prematurity, LBW, and SGA [66]. Prevention and early diagnosis are key to reduce morbidity and mortality of mother and baby.

Fetal exposure to drugs

The timing of in utero exposure to drugs plays a critical role in teratogenicity. The embryonic period (up to 10 weeks of gestation) is the most vulnerable, as organogenesis occurs during this time. While many drugs are implicated in overall teratogenicity, we discuss here the drugs that increase the risk of SVN or affect the development of the kidney. Patients with autoimmune diseases or with transplants may require treatment with teratogenic drugs like mycophenolate mofetil, cyclophosphamide, and others. Exposure to these drugs in the first trimester increases the risk of SVN and stillbirth in addition to developmental abnormalities of various organ systems [67]. Such exposures can be avoided if pregnancies are planned. Maternal exposure to angiotensin-converting enzyme inhibitors (ACEi) and angiotensin receptor blockers (ARBs) in the second and third trimester can result in ACE inhibitor fetopathy, resulting in renal tubular dysgenesis in the fetus, SGA, and oligohydramnios. These risks are higher with ARBs than with ACE inhibitors [68, 69]. There is also a high risk of acute kidney injury in the neonate, as well as long-term kidney complications [68]. Non-steroidal anti-inflammatory drugs (NSAIDs) administered in early gestation can affect the development of the fetal kidneys. Later in gestation, they can be associated with oligohydramnios, preterm labor, and neonatal AKI [68]. Exposure to NSAIDs anytime during pregnancy is associated with an increased risk of CKD in the offspring [70]. Children inadvertently exposed to such medications in unplanned pregnancies require long-term follow-up of kidney health.

Access to antenatal care

As discussed above, many pregnancies in low-resource settings are still unregistered, and deliveries are not in institutions [51]. According to WHO, 41% and 11% of women in low-income and lower-middle-income countries, respectively, do not have access to appropriate reproductive and maternal health interventions [71].

Inadequate antenatal care increases the risks of fetal distress, birth asphyxia and stillbirth as well as neonatal intensive care unit (NICU) admissions [72]. Outborn babies have a higher risk of AKI [73].

Neonatal kidney health outcomes

In this section, we review the consequences on neonatal kidney health both in terms of nephron number and the increased risk of AKI and the subsequent impact of both these factors on long-term kidney outcomes.

Low nephron endowment

According to the Developmental Origin of Health and Disease (DOHaD) paradigm, the quality of fetal development shapes individual differences in the risk of chronic illness over the lifespan [74]. In SVN, a suboptimal intrauterine environment may result in altered nephrogenesis (Fig. 1). The development of nephrons begins at 9–10 weeks of gestation and continues in utero until around 36 weeks [3, 75]. Nephrogenesis may continue for a few weeks after preterm birth. However, preterm neonates have a decreased nephrogenic width, morphologically abnormal glomeruli, and increased glomerular generations, suggesting accelerated but abnormal postnatal maturation [76]. Postnatal exposure to nephrotoxic medication also modulates nephrogenesis [76]. These developmental alterations are reflected in the strong association between gestational age as well as birth weight with nephron number [77].

As hypothesized by Brenner et al. in 1988, constraints on kidney development and a reduction in nephron endowment result in an increased risk of hypertension and chronic kidney disease [78]. A lower nephron number is associated with a reduced surface filtration area, which along with programmed changes in tubular function and vascular reactivity, predispose to the development of hypertension and increased susceptibility to CKD [79]. A kidney with fewer nephrons likely undergoes long-term hyperfiltration and glomerular hypertrophy, which over time may become maladaptive and lead to a cycle of ongoing nephron loss and glomerular sclerosis [80]. Prematurity and SGA not only affect nephron number but may impact podocyte and vascular development [81]. In preterm neonates, abnormal glomeruli have a cystic morphology, with an enlarged Bowman's space and scant capillarization [82]. Fetal adversity may therefore contribute to podocyte and glomerular injury, and accelerated post-natal podocyte depletion may contribute to the risk of long-term CKD [83].

In SVNs, a congenital nephron deficit is thought to act as a "first hit" in the kidney, which alone may not be enough to lead to hypertension and CKD. Superimposition of subsequent insults, such as AKI, hypertension, diabetes, obesity or glomerulonephritis, for example, and/or exposure to external factors such as climate change, air pollution, and infections, may lead to further nephron loss and accelerate the loss of kidney function [84–87]. Consistent with this hypothesis, the association between LBW and preterm birth with the development of CKD has been well described in population studies and is increasingly acknowledged as an important risk factor for AKI and CKD [86].

Acute kidney injury

On day 1 of life, the glomerular filtration rate (GFR, derived from Amikacin clearance) was lower in preterm and LBW infants compared to normal birth weight term controls [88]. As such, these infants are at significantly increased risk of neonatal AKI [73, 89], the long-term implications of which have yet to be well described [90].

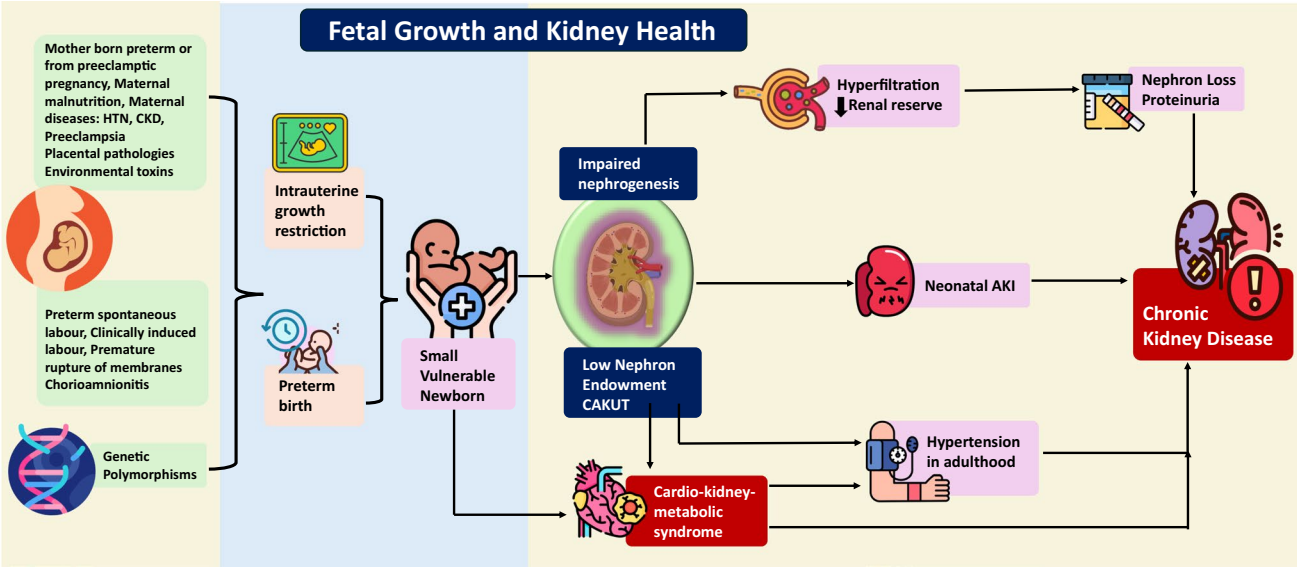

Fig. 1 Risk factors for low nephron endowment and its long-term consequences. The maternal health factors which result in the small vulnerable newborn (SVN) are highlighted. The small vulnerable newborn with a low nephron endowment and impaired nephrogenesis, compounded by the occurrence of post-natal acute kidney injury, is at risk of adverse kidney health outcomes in later life

SVNs are inherently prone to develop AKI. Newborns have a physiologically immature tubular function associated with a reduced ability to concentrate urine and reabsorb sodium, which increases the risk of prenatal AKI [91]. Prematurity also predisposes the neonate to birth asphyxia, necrotizing enterocolitis, and hemodynamically significant patent ductus arteriosus which are additional risk factors for AKI [92]. Perinatal asphyxia, an important cause of AKI in both term and preterm neonates is influenced by maternal risk factors [93].

A U-shaped distribution of AKI incidence across gestational ages was observed with extremely preterm infants having an incidence of AKI of 47.9%, compared with 18.3% in very and moderately preterm infants, and 36.7% in infants born beyond 36 weeks [94]. These observations suggest that low nephron number may enhance susceptibility to AKI in those born preterm, and highlight the exposure of SVN in the NICU to multiple simultaneous kidney stressors, including sepsis, hypotension, nephrotoxin exposures, and malnutrition [95, 96].

Increased awareness among neonatologists to minimize the risks of nephrotoxicity in neonates who have or are at risk of AKI is crucial [94]. A modified and newborn-adapted definition of AKI diagnostic criteria [94] or the rate of postnatal decline in serum creatinine [97] should be used for the diagnosis of AKI in neonates.

Neonatal AKI also negatively impacts nephrogenesis and/or leads to additional nephron loss [98]. The lower nephron number in SVN neonates, compounded by the occurrence of AKI, predisposes to hyperfiltration in remaining nephrons. Indeed, neonatal AKI is associated with features of kidney injury which develop over time—proteinuria, hypertension, and CKD [89, 99].

Concerningly, episodes of neonatal AKI are not systematically documented in discharge summaries, referrals to nephrology are not made and community pediatricians are not as aware as they should be about regularly screening these infants for kidney function and blood pressure [100, 101].

Long-term kidney outcomes

Hypertension and proteinuria

Higher blood pressure (not yet overt hypertension), proteinuria, and reduced GFR may already be evident in young children born preterm or SGA, and this risk increases with age. Among adolescent participants in the National Health and Nutrition Examination Survey, the population-attributable fraction for systolic blood pressures ≥95 th percentile and/or eGFR <90 mL/min/1.73 m^2 was 1 in 13 adolescents with LBW or 1 in 5 with very low birth weight (VLBW), respectively [80]. As these studies did not record whether individuals had neonatal AKI or not, it is unknown how this would have modulated these fractions. Still, AKI may have contributed to the greater risk of CKD among those born with VLBW. These risks and the rate of progression of proteinuria and subsequent CKD are modulated by childhood obesity [102]. This highlights the risk of excessive catch-up growth after preterm birth, which also adversely impacts cardiovascular health [103].

Maternal preeclampsia and gestational hypertension are associated with higher blood pressure in offspring during childhood, even after adjusting for adiposity [104]. IUGR may play a key role in hypertension in children born to mothers with preeclampsia, but not in those with gestational hypertension [104]. A recent meta-analysis showed systolic and diastolic blood pressure were 2 mm Hg (95%CI 1.2, 2.8) and 1.4 mm Hg (95%CI 0.9, 1.9) higher, respectively, in children born to mothers with preeclampsia [105]. SVN and maternal diabetes are also associated with an increased risk of elevated blood pressure in later life [6]. Low nephron number and the ensuing hyperfiltration lead to the cascade of chronic inflammation and nephron loss, exacerbated by AKI in the neonatal period, which may manifest early on as proteinuria and hypertension [106–108]. Indeed, a systematic review showed that LBW was associated with 1.73 times higher odds of proteinuria [7]. Hypertension and proteinuria are key determinants of the rate of progression of CKD in children as well as adults [109].

Risk of CKD

Recent studies have tried to dissect the relative impacts of LBW, preterm birth, and SGA on CKD risk. Prematurity and LBW, irrespective of the cause, have been associated with subnormal GFR in childhood and adolescence [107, 108]. The association of AKI with prematurity or LBW compounds the risk of CKD [110]. Neonates who received therapeutic hypothermia for perinatal asphyxia and developed AKI were found to have subnormal GFRs as adolescents [111], emphasizing that AKI is an important risk factor for CKD.

In a Swedish birth registry study, the risk of CKD among those born preterm was almost double that of those born at term; with the risk being highest among those born extremely preterm (HR 3.01, 95% CI 1.67 to 5.45) or who were preterm and SGA (HR 3.08, 95% CI 2.39 to 3.96) [112]. In a Norwegian birth registry, the odds ratio for CKD in those born SGA was 1.79 (95% CI: 1.65 to 1.94), and with preterm birth was 1.48 (95% CI: 1.33–1.66) [86], similar for males and females. Modulation of these risks by familial factors was evident with a higher risk of CKD among those having a sibling with LBW [113].

A fifty-year follow-up of the Norwegian cohort assessed the risk of kidney failure and reported odds ratios of 2.0 (95% CI 1.59 to 2.56) for LBW, of 1.49 (95% CI 1.27 to 1.75) for SGA, and of 1.65 (95% CI 1.31 to 2.07) for preterm birth [114]. These factors often coexist, and indeed, the investigators found that when controlling

for this, the presence of two or more birth factors was associated with an increased risk of kidney failure.

Both the Swedish and Norwegian studies also identified a risk of CKD or kidney failure among those who had been exposed to maternal preeclampsia, with risks increasing by 1.2 and 1.5-fold, respectively [112, 114]. Maternal obesity (BMI ≥ 30) was associated with a similar increased risk of offspring CKD after preterm birth [112, 114]. Exposure to maternal diabetes among adult Pima Native Americans was also associated with an incidence rate ratio of 4.12 (95% CI: 1.54 to 11.02) for kidney failure [33]. This relationship, however, dissipated when controlled for current diabetes, suggesting that developmentally programmed premature diabetes may be the predominant factor impacting CKD risk in this population.

CAKUT

CAKUT (congenital abnormalities of the kidney and urinary tract) refers to developmental abnormalities that occur due to genetic or acquired insults during embryogenesis [115]. Most CAKUT is attributed to non-genetic causes, the most important being maternal health and environment. Increased maternal age, maternal chronic health conditions [29] such as diabetes [116], hypertension, gestational conditions like preeclampsia, diabetes [116], and hypothyroidism, are associated with an increased risk of CAKUT [6, 116]. Maternal malnutrition, obesity, and deficiencies in folic acid and vitamin A [30, 117, 118] are risk factors for CAKUT. Maternal exposure to toxins including alcohol, cocaine, and drugs like ACE inhibitors affect the normal development of the kidney. Environmental factors can cause epigenetic changes influencing the development of the kidney [30, 117, 118]. While the prevalence of CAKUT is low (< 1% live births), it contributes to about 50% of CKD in children [119].

Improving maternal health and long-term kidney outcomes

Maternal health interventions

Every mother must have access to ANC under universal health care in a safe and non-judgmental environment. The World Health Organization recommends a minimum of four ANC visits [51]. Improving access to ANC through targeted interventions like conditional cash transfer programs have demonstrated improvement in birth weight-related outcomes and decreased birth weight inequalities [120]. Interventions targeting maternal health are associated with better outcomes for the mother and the offspring (Fig. 2). Maternal interventions should begin in the preconception stage and target all individuals in the reproductive age group.

Women at increased risk of poor outcomes should be identified early. Mothers who themselves were born preterm have a greater risk of pregnancy-associated hypertension and GDM as well as preterm delivery [121, 122]. These

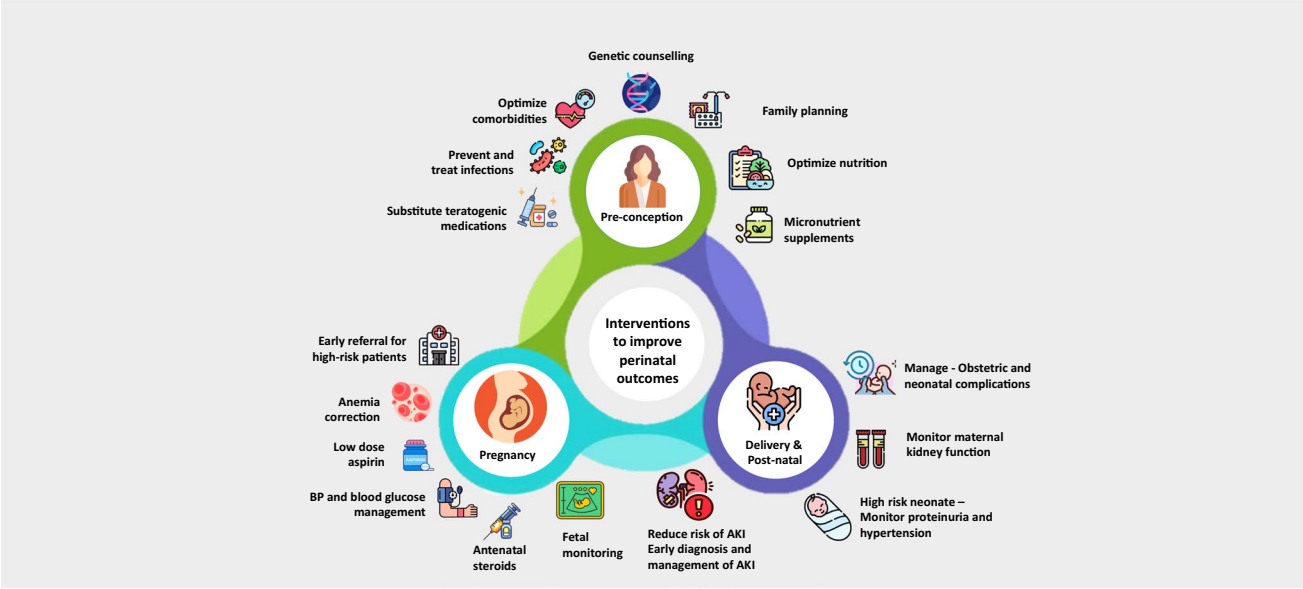

Fig. 2 Summary of the maternal, fetal, and neonatal interventions to reduce the risk of long-term consequences in small vulnerable newborns. Screening for maternal risk factors in the pre-gestational and gestational period is important to identification and early treatment. Screening of high-risk neonates and interventions to identify and reduce the risk of acute kidney injury are described. All SVNs and high-risk neonates, and their mothers, require long-term monitoring of kidney health

observations highlight the intergenerational risk of disease and morbidity that can be attenuated by optimizing maternal health in the pre-gestational period, throughout pregnancy, and by identifying and following up at-risk neonates into adulthood.

In addition to those with chronic diseases, hypertension, diabetes and obesity, pre-pregnancy counselling is required for offspring of preeclamptic pregnancies, young individuals who were born prematurely or with IUGR, or with a family history of hypertensive disorders of pregnancy to ensure adequate antenatal follow-up [35, 123, 124]. Improving nutritional status, focusing on calcium and micronutrient supplementation, and optimizing weight are important [62]. Optimum glucose control during gestation has the potential to reduce fetal complications [125]. Improving health literacy and discussing choices for family planning and birth spacing are the first steps to improving maternal health in individuals with co-morbidities. People of childbearing age who are on potential teratogenic medications must be counseled about contraception. Optimal management of chronic diseases like SLE, CKD, and organ transplants requires multidisciplinary team support throughout pregnancy.

Specialized care for the mother-baby dyad is needed when pregnancy is complicated by pre-existing conditions like CKD, the need for dialysis, following kidney transplantation, or other systemic illnesses. Mothers at high risk for preeclampsia must be started on aspirin before 16 weeks of gestation. Screening and treatment of asymptomatic bacteriuria in pregnancy has been associated with a 40% reduction in SVN [62]. Prevention and treatment of STIs like syphilis is associated with a 50% reduction in the risk of SVN [62]. Prenatal antihelminth treatment has been found to improve birth weight [126].

Early detection of fetal anomalies and fetal monitoring should be available for all high-risk mothers so that deliveries take place in settings where the babies can be managed appropriately by pediatric nephrologists and intensivists if needed.

Importantly, although not the focus here but discussed in detail elsewhere [127], adverse pregnancy outcomes such as preeclampsia increase a mother's own future risk of kidney disease and cardiovascular health, mandating counselling and long-term follow-up.

Neonatal and long-term health interventions

Given that SVNs are at risk of AKI, proteinuria, hypertension, and subsequent CKD, they require long-term follow-up and early preventive actions to preserve kidney function [128]. Clinical signs of developmental programming in the kidney in SVN infants may be detectable only in childhood or later; therefore early identification of those at risk is important. Awareness about potential long-term consequences of developmental programming and the need for follow-up should be increased among healthcare professionals and caregivers of mothers and neonates born from high-risk pregnancies. However, the need for long-term follow-up should be balanced against the risk of overdiagnosis and excessive medicalization. A careful review of the neonatal history is a first step for primary care physicians to identify small infants who require close follow-up. Particular attention should be paid to the history of maternal hypertensive disorders of pregnancy, gestational age, birth weight and weight-for-age classification, and any history of neonatal AKI.

In 2017, the LBW and Nephron Number Working Group proposed a schedule for regular monitoring of kidney function in preterm (< 32 weeks), LBW, and IUGR individuals throughout life (Table 1) [6]. The monitoring is mainly based on the measurement of BP and urine albumin whose frequency should be tailored to the patient's age, the degree of prematurity, and other neonatal risk factors (mainly AKI). Annual monitoring should start at 3 years of age (earlier in preterm newborns or cases of neonatal AKI), should be continued at 2-year intervals up to 40 years of age, and then annually thereafter. If risk factors, high BP, or proteinuria are detected, kidney function (creatinine, cystatin C) evaluation should be undertaken [128]. Early interventions targeting the control of proteinuria, blood pressure, and obesity can retard the progression of CKD [129]. Education about lifestyle and avoidance of nephrotoxins is also important for families of preterm or LBW infants. Importantly, the kidneys are not programmed in isolation. Large studies have demonstrated additional associations between SVN and subsequent hypertension, type

Table 1 Proposed schedule for regular monitoring of preterm and LBW individuals throughout life

Age	Monitoring
3 to 5 years	Annual BP measurements and urinalysis
5 to 19 years	Screening (BP and urinalysis) at well-child checks, medical visits or at 2 year intervals
19-40 years	BP, urinalysis and BMI every 2 years
> 40 years	BP, urinalysis and BMI annually

Modified from Luyckx VA et al Lancet 2017:390:424–428

BP, blood pressure; *BMI*, body mass index; *GA*, gestational age; *AKI*, acute kidney injury

*When possible, a baseline kidney ultrasound should be done to detect small kidneys, asymmetry or structural abnormalities

**In very preterm neonates (GA < 32 weeks) or in case of neonatal AKI, screening should be initiated before the age of 1 year

***If risk factors are detected (high BP, proteinuria, previous AKI, kidney anomalies, cardiovascular disease, obesity or diabetes) kidney function evaluation should be done every two years

****In low resource settings, simplified screening could coincide with public health interventions or could be undertaken by community health workers

Table 2 Summary of the maternal risk factors and consequences on fetal/neonatal and long-term kidney health

Maternal risk factors	Fetal /neonatal consequences	Long term consequences	Intervention
Pre-gestational			
Teenage pregnancy	SVN	Cardio-kidney -metabolic syndrome, CKD	Sexual health education for adolescents
Late pregnancy (> 35 years)	SVN		Antenatal counseling
High BMI	Macrosomia, prematurity, perinatal asphyxia, AKI	CKD	Nutrition and weight management
Underweight	SVN		Providing balanced energy and protein supplements decreased SGA [OR 0·71 (0·54–0·94] LBW [OR 0·60 (0·41–0·86]
Micronutrient deficiency (Folate, vitamin A)	SVN Smaller kidney volume (low nephron number)	Proteinuria, CKD	Nutritional supplementation for individuals of reproductive age reduces SVN [OR 0.9 (0.84–0.96)]
Smoking or passive smoking or tobacco consumption	SVN	Obesity, metabolic syndrome	Health education
Diabetes	Macrosomia, fetal malformations, prematurity, perinatal mortality	CKD	Glucose monitoring and control reduced Malformations [RR 0.29 (0.21–0.4] Prematurity [RR 0.85 (0.73, 0.99] SGA [RR 0.52 (0.37–0.75]
Chronic kidney disease	SVN, perinatal mortality	CKD	Multidisciplinary team support
SLE	SVN, heart blocks		Optimise disease control, plan pregnancy while in remission
Infections - Syphilis	SVN, perinatal mortality		Treatment of infection Screening and treatment of syphilis reduces SVN [OR 0·48 (0·39–0·58]
Low socioeconomic status	SVN		ANC under universal health coverage
Gestational factors			
Preeclampsia	SVN, perinatal asphyxia, neonatal AKI	Cardio-kidney-metabolic syndrome	Early diagnosis, optimal management, timely referral. Low dose aspirin decreased prematurity [OR 0·89 (0·81–0·98]
Diabetes	Macrosomia, genito-urinary malformations, hypoglycemia, prematurity	Obesity, diabetes, cardiovascular disease, CKD	Good glycemic control reduced
Infections - Asymptomatic bacteriuria	Congenital anomalies SVN		Screening and treatment of asymptomatic bacteriuria reduced SVN [OR 0·63 (0·45–0·90]
AKI	SVN		Supportive management
Exposure to drugs - ACE inhibitors NSAIDs	Renal tubular dysgenesis Prematurity, AKI	CKD	Health education, cautious use in reproductive age group
Access to antenatal care	SVN		ANC under universal health coverage

SVN, small vulnerable newborn; *AKI*, acute kidney injury; *CKD*, chronic kidney disease; *SLE*, systemic lupus erythematosus; *BMI*, body mass index; *SGA*, small for gestational age; *ANC*, antenatal care; *ACE*, angiotensin converting enzyme; *NSAIDs*, non-steroidal anti-inflammatory drugs; *OR*, odds ratio; *RR*, relative risk

The table summarises the pre-gestational and gestational risk factors for SVN and long-term adverse kidney outcomes and provides an evidence-based review of interventions that have been shown to reduce the risk of SVN and its consequences

1 and type 2 diabetes, lipid disorders, cardiac disease, and all-cause mortality. Awareness of the impact of maternal health and SVN on long-term kidney outcomes as well as the transgenerational perpetuation of disease risk must extend beyond nephrologists and neonatologists into other specialties and the community. A summary of the maternal risk factors, neonatal and long-term kidney consequences, and potential interventions is given in Table 2.

Conclusion

Advocating for a healthy mother living in a healthy society is the first step to reducing the risk of SVN. Those mothers and infants at risk must be recognized early, educated about long-term risks, and provided with the opportunities to adhere to a healthy lifestyle and to access quality and affordable primary care so that any abnormalities are detected and intervened upon early, optimizing the kidney health of the mothers, their children, and the next generations.

Key summary points

- Maternal health in the pre-gestational and gestational phases, as well as nutrition, exposure to drugs and environmental toxins, infectious and non-infectious diseases, and socio-economic conditions influence the overall development of the fetus and the fetal kidney.
- The small vulnerable newborn born from an adverse intra-uterine milieu is programmed to have low nephron number and is predisposed to acute kidney injury in the neonatal period.
- Developmental programming and early insults to the kidney have far-reaching consequences with an increased risk of cardio-metabolic disease, including CKD and preeclampsia, establishing an intergenerational cycle of disease and adverse outcomes.
- Optimizing care for all individuals at risk of pregnancy-related complications and follow-up of the small vulnerable newborn and their mothers over the long term are important to mitigate the risk of chronic disease.

Multiple choice questions

Answers appear following the References.

1. The term “small vulnerable newborn"denotes
   a. Neonates born before 37 completed weeks of gestation
   b. Neonates with a birth weight < 2.5 kg
   c. Neonates whose birth weight is < 10 th centile for gestational age
   d. All of the above

2. Which of the following statements is true:
   a. Maternal age has no impact on fetal outcomes
   b. Gestational diabetes is associated with a risk of hypodysplasia and obstructive uropathy in the fetus
   c. The rates of conception are comparable in women on chronic dialysis and kidney transplantation
   d. Micronutrient deficiency in the mother does not influence the weight of the fetus or nephron number

3. Which of the following statements is not true:
   a. The developmental origin of health and disease paradigm hypothesizes that the quality of fetal development is the first hit for the development of chronic disease
   b. Premature neonates have abnormal glomerular morphology
   c. Nephron number is significantly affected by gestational age but not by birth weight
   d. Maternal malnutrition, obesity and micronutrient deficiency are associated with congenital anomalies of the kidney and urinary tract

4. Which of the following statements is true:
   a. A minimum of ten antenatal visits is recommended by the World Health Organization
   b. Active smoking but not passive exposure to smoke is associated with adverse fetal outcomes
   c. Annual monitoring of small vulnerable newborns is recommended from 10 years of age
   d. Maternal preeclampsia is associated with a higher risk of hypertension in the offspring

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00467-025-06771-9.

Declarations

Competing interests The authors have no competing interests to declare.

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Multiple choice answers: 1. d; 2. b; 3. c; 4. d.

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